

Om Bhatt

Engineering Undergraduate (Monash University)

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[LinkedIn](#) · [GitHub Portfolio](#) · [GitHub](#)

Education

Monash University

Bachelor of Engineering (Honours), Mechatronics

Clayton, VIC

Expected Grad: 2028

- **Weighted Average Mark (WAM): 79.300 (Distinction)** · **Grade Point Average (GPA): 3.400**
- **Interests:** Digital logic design, self-taught through complex Redstone circuitry (logic gates, hex/binary decoders) · High School Robotics Club (Year 9–10).

Technical Skills

- **Hardware Engineering:** Mixed-Signal PCB Design (Altium Designer, KiCad), 4-Layer Stack-Ups & Split-Plane Power, Precision Analog Front-Ends, High-Current Power Distribution, Signal Integrity & EMI Mitigation.
- **Robotics & Control:** Inverse Kinematics, PID Control Loops, PWM Modulation (18+ channels), Autonomous Navigation, Computer Vision (Object Tracking), Microcontrollers (ESP32-S3/RP2040/Arduino).
- **CAD & Prototyping:** SolidWorks (Complex Assemblies), Rapid Prototyping (FDM), DFM, Linkage & Tendon Mechanism Design.
- **Software & AI:** Python (OpenCV, Multi-Threading), C++ (Embedded Firmware), Search & Optimisation (MCTS/UCB1, Beam Search, Genetic Algorithms), Cross-Platform Mobile (iOS/Android).

Engineering Experience

Electronics Team Member

Melbourne, VIC

Monash Unmanned Aerial Systems (MUAS)

2026 – Present

- **Project Cassowary:** Leading hardware development of the Measurement & Logging Board used to characterise heavy-lift drone propulsion systems.
- **Precision Analog Design:** Engineered a 0.1%-precision front-end capturing thrust and torque at **50 kS/s**, using an INA826 instrumentation amplifier to resolve millivolt load-cell signals while rejecting EMI from **120A** motor surges.
- **Design Review:** Validated 40V-rated converters against a 4S LiPo (16.8V peak) worst case, securing a >2x safety margin while avoiding over-engineering and cutting fabrication cost.
- **Mixed-Signal PCB Design:** Routed a 4-layer board in Altium Designer isolating analog front-ends from high-current return paths, with dual-rail LMR14050 buck converters giving the RP2040 logic independent 3.3V/5V domains protected from back-EMF spikes.

Co-Founder

Melbourne, VIC

ARES Technologies (Private R&D Team)

2025 – Present

- **Overview:** Co-founded a private engineering team building autonomous robotic systems and bio-mimetic hardware from concept to working prototype.
- **Industry Recognition:** Selected to exhibit prototypes at the **Monash University DeepNeuron Tech Expo**, presenting alongside industry leaders including Microsoft, Telstra, and Blackmagic Design.
- **Project ECKO (Assistive Haptics):** Designing the 4-layer control board for a haptic-sight blindfold that renders time-of-flight distance as vibration, driving **32 haptic motor channels** from an ESP32-S3 via dual PCA9685 controllers, with a split 3V3/+5V_MOT plane isolating motor noise from the logic and 2.4 GHz radio.
- **Project JAEGER (Bio-Mimetic Robotics):** Designed an anthropomorphic robotic hand for real-time motion mimicry, engineering a dedicated high-current power bus to eliminate brownouts during simultaneous five-finger actuation and iterating tendon-driven linkages across four SolidWorks generations.
- **Project SCORPION (Autonomous Hexapod):** Built an AI-powered hexapod coordinating **18 servos** on a tripod gait from a custom Python inverse-kinematics solver, replacing failure-prone breadboard wiring with an integrated KiCad PCB that cut chassis weight by **20%**.
- **Project SENTINEL (Computer Vision Defense):** Developed a multi-threaded object tracking system decoupling OpenCV processing from motor control, predicting target paths by vector analysis for real-time turret alignment.

Student Engineer

Melbourne, VIC

Project Weirman (Monash Academic Project)

2026

- **Result:** Placed **10th out of 160 teams** in a highly competitive university-wide engineering design challenge.
- **Autonomous Navigation:** Designed and built an automatic navigation system enabling the robot to detect deviation and self-correct its course mid-run without operator intervention.
- **Reliability by Design:** Paired a stable mechanical platform with a lean, low-power electrical architecture, prioritising bulletproof control logic over unnecessary complexity to achieve 100% system reliability during final evaluations.

Software & AI Projects

AI Agent Developer

Team Project

Pokemon TCG AI Battle Challenge (Kaggle)

2026

- Built an agent that autonomously constructs and refines a competitive deck from a pool of over **18,000 cards**, trained through more than **250,000 simulated self-play battles**.
- Replaced static greedy heuristics with Monte Carlo Tree Search (UCB1) and beam search over multi-step action sequences, while a genetic algorithm evolved ~20 board-evaluation weights via self-play tournaments.

Founder

Personal Venture

Endeavour – iOS & Android App

2026 – Present

- Building a cross-platform habit-tracking app with gamified streaks, leaderboards and head-to-head challenges, designed around social accountability.